CYS 331 —Unit 6: Case Studies

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Learning Outcomes

- Apply wireless security principles to real operational systems and environments.
- Compare how mission requirements change suitable controls across different wireless technologies.
- Identify recurring risks involving availability, trust, interception, and physical infrastructure.
- Extract transferable lessons from case studies for future design and assessment.

How to Read Wireless Case Studies

- Case studies connect theory to operational environments, threats, and protective choices.
- Each system balances coverage, mobility, reliability, cost, and security differently.
- We evaluate assets, adversaries, constraints, and failure consequences in context.
- Good analysis asks why a control fits one environment but not another.

Terrestrial Microwave Relay Systems

- Terrestrial microwave relays provide long-distance point-to-point backbone connectivity.
- Security priorities include line-of-sight protection, tower access, and backhaul availability.
- Attackers may target infrastructure, interception opportunities, or physical sabotage.
- Redundancy, encryption, and strong site security are essential controls.

Public Safety Wireless Networks

- Public safety networks support emergency responders under stressful and unpredictable conditions.
- Availability, interoperability, and resilient coverage often outweigh convenience features.
- Security must protect sensitive communications without delaying urgent operational use.
- Governance, prioritization, and incident recovery planning are especially important.

Military Tactical Radio Systems

- Military tactical radio systems operate in hostile environments with active adversaries.
- They require anti-jamming resilience, low probability of detection, and key discipline.
- Mobility, contested spectrum, and rapid mission change complicate secure design.
- Security decisions must support survivability, command integrity, and controlled access.

Satellite Communication Systems

- Satellite systems extend connectivity to remote regions beyond terrestrial infrastructure.
- Long signal paths affect latency, interception risk, and availability planning.
- Ground stations, terminals, and control links become major security targets.
- Encryption and strong control of endpoints remain critical protections.

Wide Area Wireless Data Services

- Wide area wireless data services connect distributed users across carrier-based infrastructures.
- Security depends on subscriber identity, carrier architecture, and application protections.
- Legacy services reveal how partial protection leaves important gaps unaddressed.
- Organizations should add endpoint and session security above network services.

Wireless Local and Metropolitan Area Networks

- Wireless local area networks prioritize local mobility, shared spectrum, and dense access.
- Metropolitan area networks extend coverage, often serving distributed neighborhoods or cities.
- Both require careful authentication, encryption, capacity planning, and monitoring.
- Scale changes administration, interference patterns, and attack surfaces significantly.

Cross-Case Lessons

- No wireless system is secure by technology choice alone.
- Mission, scale, threat model, and environment determine appropriate protections.
- Physical security and key management repeatedly appear across very different systems.
- Layered design remains the most transferable lesson from all case studies.

Unit 6 Summary

- Case studies show wireless security must match mission and operational reality.
- Different systems emphasize availability, secrecy, reach, interoperability, or survivability differently.
- Recurring lessons include infrastructure protection, key discipline, and layered controls.
- Strong analysis explains why one design choice fits one environment specifically.